

Polygon.io vs Alpha Vantage Data Comparison

Date: January 28, 2026 **Test Symbol:** PSTG (Pure Storage) **Date Range:** 2025-12-15 to 2026-01-15

1. Stock OHLCV Data

Metric	Polygon	Alpha Vantage
Rows returned	22	22
Response time	0.14s	0.36s
Price accuracy	All 22 dates match exactly	--
Column format	Identical CSV headers	Identical CSV headers
Sort order	Ascending (oldest first)	Descending (newest first)

Sample price comparison (all 22 dates matched):

Date	Polygon Close	Alpha Vantage Close	Match
2025-12-15	69.5700	69.5700	MATCH
2025-12-16	69.7200	69.7200	MATCH
2025-12-17	66.3200	66.3200	MATCH
2025-12-18	67.0600	67.0600	MATCH
2025-12-19	69.1300	69.1300	MATCH

Winner: Polygon -- 2.5x faster, identical data.

2. Ticker News (PSTG)

Metric	Polygon	Alpha Vantage
Articles found	3	0
Response time	1.81s	4.63s
Sentiment labels	Yes (Bullish/Bearish/Neutral)	N/A (no articles)
Summaries	Yes (rich descriptions)	N/A
Sources	Motley Fool, Benzinga	--

Polygon articles returned:

1. "The S&P 500's Best Stock in 2025 May Soar in 2026..." -- *Motley Fool*, Jan 4 -- **Bullish**

2. "A Once-in-a-Decade Investment Opportunity: The 3 Best AI Stocks..." -- *Motley Fool*, Jan 4 --
Bullish
3. "This AI Infrastructure Stock Is 'Most Likely' To Be Added To S&P 500..." -- *Benzinga*, Jan 2 --
Bullish

Alpha Vantage returned zero articles for PSTG even with monthly bucketing enabled.

Winner: Polygon -- significantly better coverage for mid-cap tickers.

3. Global News

Metric	Polygon	Alpha Vantage
Articles returned	5 (limit=5)	50 (limit=50)
Response time	0.11s	0.44s
Article quality	Curated, major outlets	Mixed (includes institutional filing noise)
Ticker tagging	Yes, relevant tickers per article	Yes, via <code>ticker_sentiment</code>

Polygon sample:

- "Tesla Is Struggling -- The Global EV Market Isn't" | TSLA, BYDDY, GM, VWAGY, F
- "90% of Nvidia's Customers Now Buy This -- and It's Not GPUs" | NVDA, ANET, CSCO

Alpha Vantage sample:

- "Robeco Schweiz AG Sells 2,888 Shares of Linde PLC \$LIN" | LIN
- "Sumitomo Mitsui Trust Group Inc. Grows Holdings in eBay Inc. \$EBAY" | EBAY

Winner: Tie -- Polygon is 4x faster with higher-quality curation; Alpha Vantage provides more volume.

4. Technical Indicators

4a. SMA (50-day)

Metric	Polygon		Alpha Vantage
Data points	21		21
Response time	0.12s		0.35s
Value accuracy	Exact match on all 21 dates		--
Date	Polygon	Alpha Vantage	Match
2025-12-16	86.3724	86.3724	MATCH
2025-12-17	85.9408	85.9408	MATCH
2025-12-18	85.4192	85.4192	MATCH

Date	Polygon	Alpha Vantage	Match
2025-12-19	84.9294	84.9294	MATCH
2025-12-22	84.4884	84.4884	MATCH

4b. RSI (14-day)

Metric	Polygon		Alpha Vantage
Data points	21		21
Response time	0.11s		0.45s
Value accuracy	~0.1% difference (negligible)		--
Date	Polygon	Alpha Vantage	Diff
2025-12-16	38.3416	38.2998	0.11%
2025-12-17	35.3289	35.2833	0.13%
2025-12-18	36.4984	36.4563	0.12%
2025-12-19	39.7791	39.7464	0.08%
2025-12-22	37.9337	37.8986	0.09%

The ~0.1% RSI difference is due to different intermediate calculation precision. Functionally identical for trading decisions.

4c. MACD

Metric	Polygon	Alpha Vantage	
Data points	21	21	
Response time	24.97s (hit rate limit)	0.43s	
Value accuracy	<0.01% difference (within rounding)	--	
Date	Polygon	Alpha Vantage	Match
2025-12-16	-4.5940	-4.5945	MATCH
2025-12-17	-4.8829	-4.8834	MATCH
2025-12-18	-4.9947	-4.9951	MATCH
2025-12-19	-4.8602	-4.8606	MATCH
2025-12-22	-4.8438	-4.8442	MATCH

Note: Polygon hit the 5 req/min rate limit on the 4th API call during this test, adding a 24s sleep. In production, the rate limiter spaces calls automatically.

Winner: Tie -- Values are identical/negligible difference. Polygon faster per-call but rate limit can add delays when making many calls in sequence.

5. Coverage Matrix

Data Type	Polygon Free Tier	Alpha Vantage Free Tier
Stock OHLCV	YES	YES
News (ticker)	YES (better mid-cap)	YES (weak mid-cap)
News (global)	YES	YES
SMA / EMA	YES	YES
RSI	YES	YES
MACD	YES	YES
Bollinger Bands	NO (auto-fallback to yfinance)	YES
ATR	NO (auto-fallback to yfinance)	YES
VWMA	NO (auto-fallback to yfinance)	NOT DIRECT
MFI	NO (auto-fallback to yfinance)	NO
Fundamentals	NO (\$199/mo)	YES
Balance Sheet	NO (\$199/mo)	YES
Income Statement	NO (\$199/mo)	YES
Cash Flow	NO (\$199/mo)	YES
Insider Transactions	NO	YES

6. Rate Limits

	Polygon Free	Alpha Vantage Free
Per minute	5 requests	5 requests
Daily cap	None	500 requests
History	2 years	Full history

Polygon's lack of a daily cap is a significant advantage when running batch analyses across many symbols.

7. Current Default Configuration

```
DATA_VENDOR_STOCK=polygon          # Faster, identical data
DATA_VENDOR_INDICATORS=polygon      # Faster, auto-fallback for unsupported
(boll, atr)
DATA_VENDOR_NEWS=polygon            # Better mid-cap coverage, faster
DATA_VENDOR_FUNDAMENTALS=alpha_vantage # Only free option for fundamentals
SOCIAL_SENTIMENT_VENDOR=aggregated  # Combines Stocktwits + ApeWisdom +
available APIs
```

8. Conclusion

The current defaults are optimal:

- **Polygon** for stock data, news, and indicators -- faster response times, no daily rate limit cap, better news coverage for mid/small-cap tickers
- **Alpha Vantage** for fundamentals -- Polygon requires \$199/mo for balance sheets, income statements, and cash flow data
- **Automatic fallback** handles edge cases: unsupported Polygon indicators (Bollinger, ATR) automatically fall back to yfinance/stockstats; rate limit errors on either vendor trigger fallback to the other